

Breeti Bandyopadhyay

COMPUTER SCIENCE AND ENGINEERING

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EDUCATION

B.Tech Computer Science

2024 to 2028

Shiv Nadar University, Noida

- **CGPA: 9.25** • Pursuing a minor in Mathematics.
- Dean's List (SNU) (2024/25): ranked in the top 10% of the Computer Science cohort.

Schooling

2021 to 2024

Birla Balika Vidyapeeth, Pilani

- Class X (CBSE): **97.6%** | Class XII (CBSE): **97%**

EXPERIENCE

Full-Stack Dev Intern

May 2026 to Jul 2026

Tech Mahindra, NSEZ Noida

- Engineered a **multi-agent LLM system for automotive warranty claims**: structured-output (Pydantic function-calling) decisioning across intake, coverage, fraud, and recommendation agents on a LangGraph pipeline with human-in-the-loop approval.
- Built a bounded conversational intake agent with conditional vision-evidence requests; tuned prompts and structured decision logic across fast/standard model tiers (Groq / DeepSeek).

Research Intern

Jun 2025 to Jul 2025

5G Use Case Lab, BITS Pilani

- Built **CNN models for radio-signal modulation classification** on the DeepSig *RadioML* 2018.01A dataset, using FFT-based spectral features as model input.
- Collaborated with EEE faculty on deep learning for wireless communication research.

RESEARCH

Research Collaborator

Dec 2024 to Jan 2025

Physics Department, BITS Pilani

- Co-authored a DFT study on reversible hydrogen storage in 2D holey graphyne, published in *Int. J. Hydrogen Energy* (Impact Factor: 8.3), DOI: <https://doi.org/10.1016/j.ijhydene.2025.150044>
- Conducted VASP simulations for structure optimization and electronic analysis (DOS, band structure); evaluated metrics meeting the DOE target.

PROJECTS

Hephaestus, an Iterative Code Performance Optimiser

- Profiles binaries with Linux perf, feeds hardware counters (IPC, cache misses, branch mispredictions) to an LLM, and applies one optimisation at a time in a loop until code reaches within 5% of theoretical CPU peak.
- Weighted scoring (70% runtime / 20% IPC / 10% cache miss rate) auto-rolls back ineffective changes; supports x86_64, arm64, wasm32, riscv64 via Docker with a Go HTTP server + Python perf agent backend and SvelteKit + Supabase frontend with run history and ZIP export.

Warranty Claims Decision Engine: Automates the warranty-claim decision with a pipeline of agents that reads the complaint, checks coverage, estimates repair cost, and screens for fraud, escalating risky or high-value claims to a manager.

- Kept payouts auditable by computing coverage and cost in plain code against a parts-and-policy database while the model only classifies and explains, so routine claims clear automatically and the rest reach a manager with their reasoning.
- **Stack:** *Python, FastAPI, LangGraph, SQLAlchemy, Next.js / TypeScript, Groq / DeepSeek*

COURSES AND CERTIFICATIONS

- **LinkedIn CoachIn Program (2026):** Selected among the top 100 women engineering students nationwide for intensive mentorship in DSA, problem-solving, and tech career prep.
- Advanced Learning Algorithms • Supervised Machine Learning: Regression and Classification • DeepLearning.AI & Stanford University.

TECHNICAL SKILLS & INTERESTS

- **Languages & AI:** Python, Java, C • FastAPI, LangGraph, SQLAlchemy, LLM agent workflows
- **ML / CV & Tools:** Scikit-Learn, TensorFlow, MediaPipe, OpenCV • Git, Linux, VS Code, VESTA, VASP

HONOURS AND ACHIEVEMENTS

- **State Topper, Rajasthan:** KAMP NASTA 2023, 1st in state (500,000+ participants).
- **School Rank 2:** Class XII (CBSE).